

# Introduction to Offensive Security

## Ethical Hacking — Course Orientation

Welcome! Let's talk about what this course is, what it isn't, and the one rule that matters most.

# Part 1

Welcome & what this is

# What is "offensive security"?

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- Testing systems by **attacking them — with permission** — to find weaknesses first.
- Also called **ethical hacking** or **penetration testing**.
- The goal is always to **help the owner get safer**.

We learn how attackers think so we can become better **defenders**.

# Say it simply

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- A locksmith who tests your locks **at your request** to show you which ones fail.
- They report what they found — they don't rob the house.
- That request and report are everything.

# Offense serves defense

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- **You can't protect what you don't understand.**
- Defenders who think like attackers spot problems sooner.
- Every finding becomes a fix.

# What this course IS

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- A from-zero intro — no experience needed.
- Hands-on, in **safe practice environments**.
- A foundation for a real cybersecurity career.

# What this course is NOT

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-   A license to hack people, schools, or websites.
-   A place to "get back at" anyone.
-   About causing damage or showing off.

Break things **in the lab**, to learn — never in the real world.

# Who you'll become

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- Someone who can set up and use an attack workstation.
- Someone who can scan, find, and explain weaknesses.
- Someone who writes a clear, professional report.
- Someone employers and colleges take seriously.

# Check your understanding

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In one sentence: **what makes this "ethical" hacking** instead of just hacking?

# Answer

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- **Permission.** You only test what you own or are **authorized** to test.
- Same tools, same skills — the difference is **consent and intent**.

# Part 2

## The hacker mindset

# The hacker mindset

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- Relentless curiosity: *"How does this really work?"*
- Willingness to try, fail, and try again.
- Attention to small details others skip.

# Curiosity, channeled

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- The same drive can **build** or **break**.
- This course points it at building safer systems.
- Curiosity is a strength — direction is a choice.

# "Try harder"

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- Security work means getting stuck — a lot.
- Progress comes from patience and notes, not luck.
- Being stuck is normal; quitting is the only failure.

Document what you tried. Then try one more thing.

# Two paths, same skills

| Path  | Outcome                            |
|-------|------------------------------------|
| Build | Better defenses, a career, respect |
| Break | Harm, charges, a closed future     |

The skills are identical. **You** choose the path.

# Hats: who's who

| Hat                                                                                                | Who they are                                 |
|----------------------------------------------------------------------------------------------------|----------------------------------------------|
|  <b>White hat</b> | Hacks legally, with permission, to help      |
|  <b>Black hat</b> | Criminal — no permission, causes harm        |
|  <b>Gray hat</b>  | No clear permission, no malice — still risky |

# Red team vs blue team

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- **Red team** = the attackers (us, in this course).
- **Blue team** = the defenders who detect and stop them.
- Great defenders study offense; great attackers respect defense.

# Check your understanding

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A "gray hat" hacks a company **without asking** but reports the bug and asks for nothing. Are they safe from the law?

# Answer

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- **No.** No permission = still likely illegal.
- Good intentions don't replace **authorization**.
- The professional path is to ask first (or use a bug-bounty program).

# Part 3

The law & ethics — the heart of this course



# The one rule

Authorization is the **ONLY** line between a penetration tester and a criminal.

You may attack **only** systems you **own** or have **explicit written permission** to test.

# What is "scope"?

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- **Scope** = exactly what you're allowed to test, in writing.
- It names the targets, the methods, and the time window.
- Outside the scope = **not allowed**, period.

# In this class, the scope is...

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-  The practice targets your teacher provides.
-  Pre-authorized platforms (TryHackMe, picoCTF...).
-  Everything else — always.

# The CFAA

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- **Computer Fraud and Abuse Act** — the main U.S. anti-hacking law.
- Bans accessing computers **without authorization**.
- Penalties range from fines to serious felonies.

# State laws too

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- Every U.S. state has its own computer-crime laws.
- Many are as strict as — or stricter than — the CFAA.
- "I didn't know" is not a defense.

# Minors are NOT exempt

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- Students can face **juvenile or criminal charges**.
- Plus school discipline and lasting consequences.
- Colleges and employers can see the fallout for years.

This is real. Treat it that way.

# Even "just looking" can be illegal

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- Unauthorized **scanning** can break the law.
- So can opening an account or file that isn't yours.
- Testing a friend's or the school's systems counts.

# Tools are dual-use

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- The tools we use are used by pros **and** criminals.
- Owning them is generally fine; **unauthorized use** is not.
- The tool is neutral — your authorization is what matters.

# Discuss: where's the line?

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You notice the school website has an obvious flaw. You're "pretty sure" you could get in.

- Is testing it okay? Why or why not?
- What *should* you do instead?

# Responsible disclosure

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- Found a real flaw by accident? **Don't exploit or share it.**
- Report it **privately** to the owner (or your teacher).
- Let them fix it before anyone else learns of it.

# Bug bounties

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- Many companies **pay** researchers to report flaws legally.
- It's a real career and a legal way to use these skills.
- Programs define a clear, written **scope** — just like a pentest.

# The habit that defines a pro

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Report, don't exploit.

- Reporting earns trust, money, and a reputation.
- Exploiting earns a record.

# Check your understanding

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You find a way into a stranger's account on a game you play. What's the right move?

# Answer

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- **Do not log in.** That's unauthorized access.
- Report it to the game's security/contact page.
- If unsure, tell a trusted adult or your teacher.

# Part 4

How the course works

# The big picture

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- **6 modules → 18 units**, about one unit per week.
- We build from foundations up to a full capstone.
- Every unit pairs an idea with a **hands-on lab**.

# Your weekly rhythm

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- Learn the concept.
- Practice it in a guided lab.
- Write it up in your **lab journal**.

# The lab journal

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- Your running record of every lab.
- For each: goal, tools, commands, results, reflection.
- It models how real pentesters take notes — and it's graded.

If it isn't written down, it didn't happen.

# Roadmap — Modules 0-1

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- **0 · Foundations & ethics** → build a safe lab.
- **1 · Technical foundations** → networking, Linux, Bash, Python.

# Roadmap — Modules 2-3

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- **2 · Reconnaissance** → find information, then scan.
- **3 · Exploitation** → web attacks, SQLi, exploits, passwords.

# Roadmap — Modules 4-5

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- **4 · Post-exploitation** → privilege escalation, Metasploit.
- **5 · Putting it together** → reporting, capstone CTF, careers.

# Tools you'll meet

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- **Kali Linux** — the attack workstation.
- **nmap, Wireshark** — scanning and traffic.
- **Burp Suite, Metasploit** — web and exploitation.

All used **only** against safe practice targets.

# Check your understanding

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What three things happen every week in this course?

# Answer

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1. **Learn** the concept.
2. **Practice** it in a hands-on lab.
3. **Document** it in your lab journal.

# Part 5

Your lab — safe and isolated

# Three ways to practice

| Platform              | What it is                     |
|-----------------------|--------------------------------|
| TryHackMe / HTB       | Guided, browser-based rooms    |
| picoCTF / OverTheWire | Free CTF & command-line games  |
| Local VMs             | Kali + targets on your machine |

# Browser platforms

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- **TryHackMe / HTB Academy** — guided, low-setup.
- Run in a browser; great on most school hardware.
- The targets are built to be attacked — legally.

# CTF & wargames

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- **picoCTF** — beginner capture-the-flag challenges.
- **OverTheWire** — Linux command-line wargames.
- Free, safe, and pre-authorized to practice on.

# Local virtual machines

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- A **virtual machine** is a computer running inside your computer.
- **Kali** is the attacker; a vulnerable VM is the target.
- If a VM breaks, you reset it — no harm done.



# The isolation rule

Lab VMs stay on an isolated network — never the school network or open internet while attacking.

# Why isolation matters

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- **Technical:** keeps attacks from leaking out.
- **Ethical/legal:** guarantees you only touch in-scope targets.
- It's how we keep "lab only" actually true.

# How we prove it's isolated

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- Use a **host-only** (or internal) network in VirtualBox.
- Test: you **can** reach the target...
- ...but you **cannot** reach the real internet while attacking.

# Snapshots = undo button

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- Take a **snapshot** of a clean VM.
- "Owned" or broke it? Restore in seconds.
- Experiment freely — you can always reset.

# Lab safety rules

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- Attack **only** the provided targets.
- No malware on school machines outside the lab.
- Never use real personal data or real passwords.

# Check your understanding

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During an attack lab, your Kali VM **can** ping the internet. Is that okay?

# Answer

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- **No.** That means it isn't isolated.
- Switch to **host-only** and re-test before continuing.
- Reaching the internet means you could hit out-of-scope targets.

# Part 6

## Working & learning well

# Learn like a pro

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- Take notes you could follow again next week.
- Re-do labs you found hard until they feel easy.
- Teach a concept to a partner to test yourself.

# Getting unstuck

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- Re-read the error message — slowly.
- Check the man page or `--help`.
- Change **one** thing at a time, and note results.

# A note on using AI

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Use of Claude and ChatGPT is **encouraged** — but make sure **you** understand the content instead of blindly telling it what to do.

# AI: tutor, not autopilot

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-  Explain a concept a new way; help debug; check your thinking.
-  Do the learning for you.
- If AI does the work, **you** can't do it (or defend against it) later.

# Help vs. cheating

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- **Help:** pairing, hints, explaining ideas. 
- **Cheating:** copying a writeup as your own. 
- You keep your **own** journal and write your **own** reports.

# Part 7

## Expectations & getting started

# Before you touch a tool

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- Read and **sign the Acceptable Use & Ethics Agreement.**
- Yes — a **parent/guardian** signs too.
- No signature, no lab. No exceptions.

# What I expect from you

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- Stay in scope, every single time.
- Document your work honestly.
- Report problems instead of exploiting them.

# What you can expect from me

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- Clear labs, real tools, and honest feedback.
- A safe place to be curious and make mistakes.
- Support when you're stuck — that's the job.

# How you're graded

| Part                     | Weight |
|--------------------------|--------|
| Labs & lab journal       | 30%    |
| Quizzes & checks         | 20%    |
| Projects                 | 25%    |
| Capstone                 | 20%    |
| Professionalism & ethics | 5%     |

# Where this can lead

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- Pentester, SOC analyst, red/blue team, AppSec, GRC...
- A large, growing field with strong demand.
- Explore real roles and pay on **CyberSeek**.

# A certification roadmap

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- **Security+** — solid entry-level cert.
- **eJPT / PNPT** — hands-on next steps.
- **OSCP** — the pro pentest cert (this course's "north star").

# Check your understanding

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Name the **one rule** that everything else in this course depends on.

# Answer

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**Authorization** — only test what you own or are given written permission to test.

Everything we do bends to that rule.

# Bring with you

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- Curiosity — lots of it.
- Patience for getting stuck.
- Good judgment about right and wrong.

# Let's get started

**Unit 1:** What Is Offensive Security? Ethics, Law & the Hacker Mindset

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